# UNIT–III: Important Questions (2 Marks)

Text-only answers. Blank space is provided for drawing diagrams by hand.

### Q1. What is the main advantage of Adaptive Optimization methods over standard Stochastic Gradient Descent?

Adaptive Optimization methods automatically adjust the learning rate for each parameter.
**Advantages:**
1. They provide faster and more stable training.
2. They reduce manual learning rate adjustments.
3. They help the model learn efficiently.

Diagram space:

### Q2. What is the role of the Learning Rate in optimization? What are the consequences of setting it too high or too low?

The **learning rate** controls the size of the step taken when updating weights and biases during training.

**High Learning Rate:** Learning becomes faster, but it may overshoot the minimum and make training unstable.
**Low Learning Rate:** Learning becomes stable, but it takes more time to reach the minimum loss.

A suitable learning rate helps in fast and stable learning.

Diagram space:

### Q3. What is the Hessian Matrix and what is its role in second-order optimization?

A **Hessian Matrix** is a square matrix containing the second-order partial derivatives of a function.
**Uses:**
1. It helps to find the curvature of a loss function.
2. It is used to identify minimum, maximum, and saddle points.
3. It helps in second-order optimization.

Diagram space:

## Q4. Define Batch Normalization (BN) and state two key benefits it provides during training.

**Batch Normalization** is a technique that normalizes the values in a neural network layer for each mini-batch.
**Benefits:**
1. It improves training stability.
2. It often makes training faster.
3. It helps the network learn efficiently.

Diagram space:

## Q5. Define RMSProp optimizer and state its primary purpose.

**RMSProp (Root Mean Square Propagation)** is an optimization algorithm that adjusts the learning rate using the moving average of squared gradients.
**Purpose:**
1. It reduces large changes in weight updates.
2. It makes training more stable.
3. It helps the model learn efficiently.

Diagram space:

## Q6. What are the First-Moment and Second-Moment estimates in the Adam optimizer?

Adam uses two moment estimates to update the weights.
1. **First Moment ($m_t$):** Moving average of gradients. It gives the direction of the update.
2. **Second Moment ($v_t$):** Moving average of squared gradients. It helps adjust the size of the update.

**Easy memory:** First = Gradient average; Second = Squared-gradient average.

Diagram space:

## Q7. What is Dropout in Neural Networks? What is the role of the dropout retention probability hyperparameter p?

**Dropout** is a regularization technique that randomly turns off some neurons during training to reduce overfitting.

**Retention Probability (p):**

1. It represents the probability of keeping a neuron active.
2. A neuron is kept with probability p and dropped with probability 1-p.
3. It helps prevent overfitting.

Diagram space:

## Q8. What is DropConnect? How does it differ conceptually from Dropout?

**DropConnect** is a regularization technique that randomly removes some weights (connections) during training.

**Difference:**

**Dropout:** Randomly removes neurons. Reduces neuron outputs.

**DropConnect:** Randomly removes connections (weights). Sets selected weights to zero.

Both techniques help reduce overfitting.

Diagram space:

## Q9. Define the Saddle Point problem in deep network loss surfaces.

A **Saddle Point** is a point on a loss surface where the gradient is zero, but it is neither a minimum nor a maximum.

- It has upward curvature in some directions.
- It has downward curvature in other directions.
- It makes optimization difficult.

Diagram space:

## Q10. Explain the concept of Momentum in Gradient Descent and how it dampens oscillations.

**Momentum** is an optimization technique that uses previous gradients along with the current gradient to update weights.

**Uses:**

1. It speeds up learning in the correct direction.
2. It reduces unnecessary oscillations.
3. It helps the model reach the minimum loss faster.

Diagram space:

## Q11. List out various optimization methods used in deep learning in chronological order of their evolution.

The commonly taught evolution of optimization methods is:

1. Gradient Descent (GD)
2. Stochastic Gradient Descent (SGD)
3. Momentum
4. Nesterov Accelerated Gradient (NAG)
5. AdaGrad
6. RMSProp
7. Adam.

Diagram space: